

## Polynomial Functions Quiz

Algebra 2 Version A

Name: \_\_\_\_\_

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**1. (4 pts)** Factor completely:  $2x^2 - 2x - 12$

**2. (8 pts)** Let  $f(x) = x^2 + 3x - 4$ . Use algebra to answer the following.

**a.** Find all real zeros of  $f$ .

**b.** Find the  $y$ -intercept.

**c.** Describe the end behavior of  $f$ .

**d.** State the interval(s) where  $f(x) < 0$ .

**3. (10 pts)** Let  $f(x) = x^3 - 5x^2 + 2x + 8$ . Use algebra to answer the following.

**a.** Find all real zeros of  $f$ .

**b.** Find the  $y$ -intercept.

**c.** Describe the end behavior of  $f$ .

**d.** State the interval(s) where  $f(x) > 0$ .

**e.** What is the maximum number of turning points  $f$  can have?

**4. (12 pts)** Let  $f(x) = 8 - 2(x - 3)^2$ , written in vertex form  $a(x - h)^2 + k$ .

**a.** Describe the three transformations applied to  $y = x^2$  to produce  $f$ .

**i.** Horizontal shift: right 3 units.

**ii.** Vertical stretch/reflection by a factor of 2 with a reflection over the  $x$ -axis.

**iii.** Vertical shift up 8 units.

**b.** Find the key features of  $f$ . Note how sub-parts i–iv are arranged in a two-column grid.

**i.** State the vertex.

**ii.** State the axis of symmetry.

**iii.** Find the  $x$ -intercepts algebraically.

**iv.** Find the  $y$ -intercept.

**c.** State the range of  $f$  using interval notation.